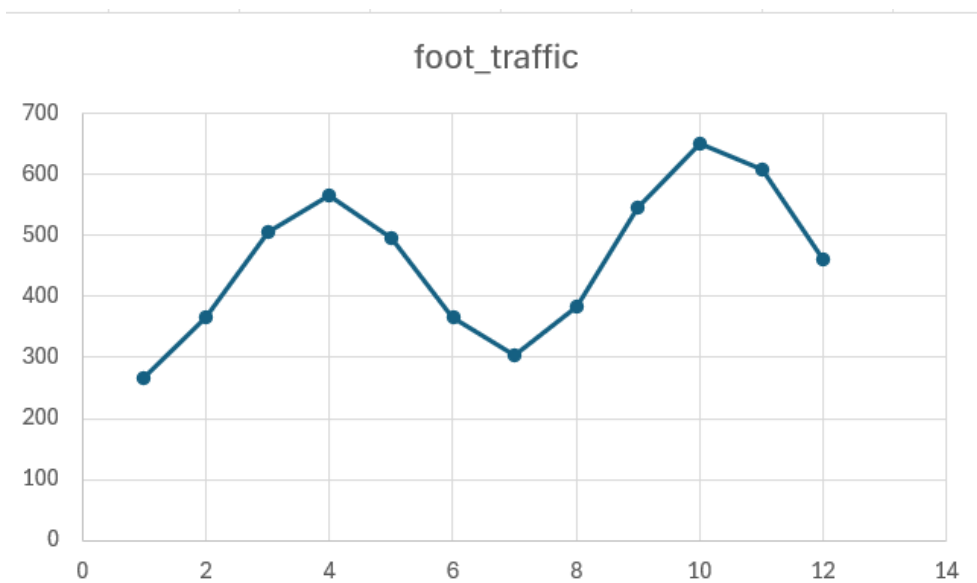


Module 08 – Scheduling Problem

Exploratory Data Analysis

Mean employee salary monthly: \$6,614.99

Agencies	Months off	Salary
Lickety Split Co.	1,2,3,4,7,8,9,10,11,12	\$15,358
The Whipped Wish	1,2,3,4,8,9,10,11,12	\$21,015
Snickerdoodle Street	1,2,3,4,5,6,7,11,12	\$25,302
Toffee Twirl	1,5,6,7,8,9,10,11,12	\$28,449
Zesty Zagnut	1,2,3,4,5,6,7,8,9,10	\$16,282
Caramel Carousel	3,4,5,6,7,8,9,10,11	\$20,544
Full time employees	0	\$79,380



Based on this foot traffic chart, there are a lot higher traffic trends in the spring and fall months and less traffic in the summer months.

Model Formulation

X_1 = scheduled workers assigned at Lickety Split co.

X_2 = scheduled workers assigned at The Whipped Wish

X_3 = scheduled workers assigned at Snickerdoodle Street

X_4 = scheduled workers assigned at Toffee Twirl

X_5 = scheduled workers assigned at Zesty Zagnut

X_6 = scheduled workers assigned at Caramel Carousel

X_7 = full-time employees

Minimize the total wage expense

$$\text{MIN} = 15,358X_1 + 21,015X_2 + 25,302X_3 + 28,449X_4 + 16,282X_5 + 20,544X_6 + 79,380X_7$$

Workers required each month

01: $0X1+0X2+0X3+0X4+0X5+1X6+1X7 \geq 267$
 02: $0X1+0X2+0X3+1X4+0X5+1X6+1X7 \geq 366$
 03: $0X1+0X2+0X3+1X4+0X5+0X6+1X7 \geq 505$
 04: $0X1+0X2+0X3+1X4+0X5+0X6+1X7 \geq 565$
 05: $1X1+1X2+0X3+0X4+0X5+0X6+1X7 \geq 495$
 06: $1X1+2X2+0X3+0X4+0X5+0X6+1X7 \geq 365$
 07: $0X1+1X2+0X3+0X4+0X5+0X6+1X7 \geq 304$
 08: $0X1+0X2+1X3+0X4+0X5+0X6+1X7 \geq 384$
 09: $0X1+0X2+1X3+0X4+0X5+0X6+1X7 \geq 545$
 10: $0X1+0X2+1X3+0X4+0X5+0X6+1X7 \geq 651$
 11: $0X1+0X2+0X3+0X4+1X5+0X6+1X7 \geq 608$
 12: $0X1+0X2+0X3+0X4+1X5+1X6+1X7 \geq 461$
 $X_i \geq 0$ and integer for all i

Model Optimized for Min Costs to Cover Store Foot Traffic

Agencies	Days On = 1, Days Off = 0												Workers Scheduled	Wages per Worker
	1	2	3	4	5	6	7	8	9	10	11	12		
Lickety Split Co.	0	0	0	0	1	1	0	0	0	0	0	0	0	\$ 15,358
The Whipped Wish	0	0	0	0	1	1	1	0	0	0	0	0	0	\$ 21,015
Snickerdoodle Street	0	0	0	0	0	0	0	1	1	1	0	0	0	\$ 25,302
Toffee Twirl	0	1	1	1	0	0	0	0	0	0	0	0	0	\$ 28,449
Zesty Zagnut	0	0	0	0	0	0	0	0	0	0	1	1	0	\$ 16,282
Caramel Carousel	1	1	0	0	0	0	0	0	0	0	0	1	1	\$ 20,544
Full time employees	1	1	1	1	1	1	1	1	1	1	1	1	1	\$ 79,380
Available	495	565	565	565	495	495	495	651	651	651	608	608		
Required	267	366	505	565	495	365	304	384	545	651	608	461	Total ->	\$ 47,071,494

This model is representing the amount of workforce needed each month at each agency while minimizing total workforce production costs.

Model with Stipulation

1. Add a constraint to your model that takes your first model's recommended number of full-time employees and constrains it to be only 80% of it. Add a text explanation of the change in the optimal value as well as any other changes noticed between the models.

Agencies	Days On = 1, Days Off = 0												Workers Scheduled	Wages per Worker
	1	2	3	4	5	6	7	8	9	10	11	12		
Lickety Split Co.	0	0	0	0	1	1	0	0	0	0	0	0	99	\$ 15,358
The Whipped Wish	0	0	0	0	1	1	1	0	0	0	0	0	0	\$ 21,015
Snickerdoodle Street	0	0	0	0	0	0	0	1	1	1	0	0	255	\$ 25,302
Toffee Twirl	0	1	1	1	0	0	0	0	0	0	0	0	169	\$ 28,449
Zesty Zagnut	0	0	0	0	0	0	0	0	0	0	1	1	212	\$ 16,282
Caramel Carousel	1	1	0	0	0	0	0	0	0	0	0	1	0	\$ 20,544
Full time employees	1	1	1	1	1	1	1	1	1	1	1	1	396	\$ 79,380
Available	396	565	565	565	495	495	396	651	651	651	608	608		
Required	267	366	505	565	495	365	304	384	545	651	608	461	Total ->	\$ 47,666,586

The optimal value decreased compared to the original model as well as the amount of scheduled workers at each month per agency.

2. Convert your model (or do the math out yourself) to figure out what monthly salary you would need to pay your full-time employees to only have full-time workers at the same optimal cost as the original model.

Highest daily requirement is 651 workers

Exchange the current workforce (495) full-time employees to new workforce of 651 full-time employees

Keeping the total annual cost at \$47,071,494

Annual salary per employee: $\$47,071,494 / 651 = \$72,314$

Monthly salary per employee: $\$47,071,494/12 = \$6,026$

3. Considering trends and seasonality of this business, what would you recommend leadership to do? Feel free to play with the model and recommend something else.

Considering the trends of this business, I would recommend leadership to optimize full-time and temporary workforce balance and instead of eliminating temporary workers completely, they could maintain a hybrid workforce and increase the number of full-time employees from 495 to around 550-575 and use temporary workers only for peak periods, reducing overall labor costs by avoiding excess full-time wages in low-demand periods.